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Extended Hybrid Technique for Control Redesign with Stabilization and Correction

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Abstract— **The Exact Model Matching (EMM) problem knows an Approximate Model Matching (AMM) solution in the Fault Tolerant Control (FTC) context, solution obtained using either the Modified Pseudo-Inverse Method (MPIM) or the Classical Robust Optimal Model Matching (CROMM) technique. In this paper, the authors reiterate the AMM solution with MPIM-CROMM and propose another hybrid technique of MPIM with Generalized Linear Quadratic Regulator (GLQR) stabilization and control correction using one of the ROMM variants, called the Extended approach (EROMM). The specificity of the MPIM-GLQR-EROMM as an AMM solution for FTC is also shown in its application on the B747 short-period model impaired with two structural and actuator faults.**

I. Introduction

In the context of Fault Tolerant Control (FTC), the Model Matching (MM) problem rarely encounters an exact (EMM) solution but an approximate (AMM) one is often preferred [1], [2]. Among the solutions existing in the literature for the AMM problem, the authors brought contributions to the Modified Pseudo-Inverse Method (MPIM) [3] and to the Classical Robust Optimal Model Matching (CROMM) approach [4] in [5]-[6].

In this paper, the authors develop the hybrid technique of MPIM with ROMM and continue the efforts from [5]-[6], relating the previous results from yet another perspective. In the first step of system stabilization, the Generalized Linear Quadratic Regulator (GLQR) replaces the simple LQR used in the original method, as the GLQR possesses the supplementary intrinsic stability degree property, with direct influence on the performance of the natural modes of system behavior. For the quadratic stabilization, the weighting matrices from the new Extended Robust Optimal Model Matching (EROMM) are used, action whose effect is visible also in the second step of control correction, showing important differences from the previous result of the CROMM use published in [5]-[6].

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The paper is organized as follows: Section II introduces the EMM problem with the Pseudo-Inverse Method (PIM) solution, while Section III addresses the AMM problem with the Modified PIM (MPIM) solution. The hybrid technique of MPIM with GLQR and EROMM is developed in Section IV, and an illustrative application example of the MPIM-GLQR-EROMM for an impaired and unstable B747 short-period model is provided in Section V. The paper is concluded in Section VI, while, in Appendix I, the ROMM approaches to be proposed in another paper are introduced briefly for completeness.

II. Exact Model Matching - Pseudo-Inverse Method

A. Nominal Operation

Consider the continuous linear time-invariant (LTI) deterministic system whose nominal operation is modeled by

$$M_n : \dot{x}(t) = A_n x(t) + B_n u_n^*(t), \text{ with } x(0) = x_0, \qquad (1)$$

where $A_n \in \mathbb{R}^{n\times n}$ and $B_n \in \mathbb{R}^{n\times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u_n^*(t) \in \mathbb{R}^m$ are the state and control vectors, and $x_0 \in \mathbb{R}^n$ is the initial condition.

In nominal operation, the linear state-feedback control law

$$u_n^*(t) = -K_n^* x(t), \qquad (2)$$

leads to the closed-loop system

$$\hat{M}_n^* : \dot{x}(t) = \hat{A}_n^* x(t), \text{ with } \hat{A}_n^* = A_n - B_n K_n^*, \qquad (3)$$

which is stable and provides the ideal dynamic performance, where by $\hat{\alpha}_n^* \triangleq |\lambda(\hat{A}_n^*)|$ the maximum absolute value of the real-part of the eigenvalues of $\hat{A}_n^*$ is denoted.

Model $\hat{M}_n^*$ is taken as the reference model whose dynamics $\hat{A}_n^*$ is to be matched in any other operating conditions different from the ideal one, while $\hat{\alpha}_n^*$ indicates the minimum distance imposed between the eigenvalues position to the imaginary axis in the complex plane, namely the Prescribed Degree of Stability (PDS) [7]; obviously, $\hat{\alpha}_n^*$ is also a crucial factor which limits the performance offered by the natural modes of system behavior.

B. Post-Fault Operation

Whether parametric faults are considered, assuming that an LTI model is still able to capture its behavior, the system to be controlled is described by the pair (A_f, B_f) provided by an on-line Fault Detection, Isolation, and Identification / Estimation (FDIE) module.

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That is, after the fault estimation time t_f, the post-fault system operation is modeled by

$$M_f : \dot{x}(t) = A_f x(t) + B_f u_f(t) . \tag{4}$$

In post-fault operation, the Exact Model Matching (EMM) technique with state-feedback aims at designing the linear control law

$$u_f(t) = -K_f x(t) , \tag{5}$$

such that the resulting closed-loop behavior matches that of the reference model $\hat{M}_n^*$ from (3) (full state is assumed to be available and known after the fault - otherwise, state estimation must be performed before controller redesign).

The immediate solution to the EMM problem is obtained by solving the matrix equation

$$\hat{A}_f \triangleq A_f - B_f K_f = \hat{A}_n^* , \tag{6}$$

whose necessary and sufficient condition for a solution to exist is expressed as (see [8])

$$\mathrm{Im}\left(A_f - \hat{A}_n^*\right) \subseteq \mathrm{Im}\left(B_f\right) , \tag{7}$$

and the solution is given by

$$(K_f^{\mathrm{PIM}} \triangleq) K_f = B_f^{\dagger}(A_f - \hat{A}_n^*) , \tag{8}$$

where $(.)^{\dagger}$ stands for the pseudo-inverse matrix.

C. Pseudo-Inverse Method

As EMM is often impossible $\left(\hat{A}_f \neq \hat{A}_n^*\right)$, an approximate control solution may be computed as $K_f = \arg\min_{K_f} J_f(K_f)$, by minimizing the criterion (see [3])

$$J_f(K_f) = \|\hat{E}_f\|_F , \tag{9}$$

with the impairment matrix

$$\hat{E}_f = \hat{A}_n^* - (A_f - B_f K_f) \; (\Leftrightarrow \hat{A}_f = \hat{A}_n^* - \hat{E}_f) , \tag{10}$$

where $\|.\|_F$ is the Frobenius matrix-norm; unfortunately, for certain fault cases, PIM encounters instability problems (see, e.g., [9]).

III. Approximate Model Matching I - Modified Pseudo-Inverse Method

A. Modified Pseudo-Inverse Method without First Step Stabilization

The improved PIM, namely the Modified Pseudo-Inverse Method (MPIM), restricts the computation of K_f to guarantee the stability of the post-fault system $\Lambda(\hat{A}_f) \subseteq \mathbb{C}^-$ while achieving as much of the closed-loop nominal performance as possible, that is $\min_{K_f} J_f(K_f)$, where $\Lambda(.)$ denotes the spectrum of the matrix-argument. It admits a closed-form solution for single-input multi-output systems (SIMO) ($B_f \in \mathbb{R}^{n\times 1}$) (or for systems decomposable into SIMO subsystems), solution which is investigated and improved in this paper.

Therefore, in MPIM for SIMO, the control gains $k_{f_j}^{\mathrm{MPIM}}$ (for $j = 1,2,\ldots,n$) computed as

$$k_{f_j}^{\mathrm{MPIM}} = k_{f_j}^{\mathrm{PIM}} , \text{ if } |k_{f_j}^{\mathrm{PIM}}| \leq \tilde{\delta}_f^{\mathrm{PIM}} , \tag{11a}$$

$$k_{f_j}^{\mathrm{MPIM}} = \mathrm{sgn}\left(k_{f_j}^{\mathrm{PIM}}\right) \tilde{\delta}_f^{\mathrm{PIM}} , \text{ otherwise} , \tag{11b}$$

are designated to solve the reconfigurable control problem with stability guarantees.

In (11), $\tilde{\delta}_f^{\mathrm{PIM}}$ represents the truncated robust stability-margin, computed as $\tilde{\delta}_f^{\mathrm{PIM}} = \delta_f^{\mathrm{PIM}} - \varepsilon_f^{\mathrm{PIM}}$, for some small rounding error $\varepsilon_f^{\mathrm{PIM}}$, and $\mathrm{sgn}(.)$ stands for the sign-function of the real-argument.

This comes from the fact that if the post-fault closed-loop system obtained with PIM, described as

$$\dot{x}(t) = \hat{A}_f^{\mathrm{PIM}} x(t) = (A_f - B_f K_f^{\mathrm{PIM}}) x(t) ,$$

is unstable, then the pseudo-inverse control must be accommodated through the modified pseudo-inverse method such that the MPIM closed-loop system, described as

$$\begin{aligned} \dot{x}(t) = \hat{A}_f^{\mathrm{MPIM}} x(t) &= (A_f - B_f K_f^{\mathrm{MPIM}}) x(t) \\ &= (A_f - \delta_f^{\mathrm{PIM}} B_f K_f^{\mathrm{PIM}}) x(t) , \end{aligned}$$

becomes stable, from where $K_f^{\mathrm{MPIM}} = f(K_f^{\mathrm{PIM}})$.

The idea is to see the closed-loop matrix in the post-fault case $\hat{A}_f^{\mathrm{MPIM}}$ as an impaired / perturbed matrix

$$\hat{A}_f^{\mathrm{MPIM}} \triangleq A_f - B_f K_f^{\mathrm{MPIM}} = A_f - E_f^{\mathrm{MPIM}} , \tag{12}$$

where the impairment matrix (which can be assimilated as a perturbation matrix, as well), denoted by E_f^{MPIM}, is taken as $B_f K_f^{\mathrm{MPIM}} = B_f f(K_f^{\mathrm{PIM}})$, and is assumed to be bounded by δ_f^{PIM}, from where it will follow that $K_f^{\mathrm{MPIM}} = \delta_f^{\mathrm{PIM}} K_f^{\mathrm{PIM}}$.

Furthermore, the condition

$$|k_{f_j}^{\mathrm{MPIM}}| \leq \tilde{\delta}_f^{\mathrm{PIM}} , \text{ for } j = 1,2,\ldots,n , \tag{13}$$

will guarantee the stability of $\hat{A}_f$ (to compute δ_f^{PIM} for E_f^{MPIM}, see [10] and [11] for the time- or frequency- domain approaches).

However, one of the major deficiencies of MPIM is that the robust stability-margin can only be used when matrix A_f remains stable after the fault; otherwise, the system must first be stabilized using another technique (like the linear quadratic regulator), and then a control correction of the stabilizing gain must be done.

B. Modified Pseudo-Inverse Method with LQR Stabilization and Correction

In such situations, for certain instability cases, in the classical paper on MPIM [3], it was proposed to firstly obtain the stabilizing gain through solving the corresponding simple Continuous Algebraic Riccati Equation (CARE) associated with the post-fault pair (A_f, B_f), that is

$$CARE_f : A_f^T X_f + X_f A_f - X_f B_f R_f^{-1} B_f^T X_f + Q_f = 0 , \tag{14}$$

where the solution to CARE_f is computed as

$$K_f^{\mathrm{CARE}} = R_f^{-1} B_f^T X_f , \tag{15}$$

and then to make a control correction as

$$K_f^{\text{MPIM-LQR}} = K_f^{\mathrm{CARE}} + \tilde{\delta}_f^{\text{PIM-LQR}} \Delta K_f^{\text{PIM-LQR}} , \tag{16}$$

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where the control accommodation difference $\Delta K_f^{\text{PIM-LQR}}$ is obtained as

$$\Delta K_f^{\text{PIM-LQR}} = B_f^{\dagger}\left(\hat{A}_f^{\text{LQR}} - \hat{A}_n^*\right), \text{ for } \hat{A}_f^{\text{LQR}} = A_f - B_f K_f^{\text{CARE}}, \tag{17}$$

and $\delta_f^{\text{PIM-LQR}}$ is computed accordingly.

However, for the first step of stabilization, solving the CARE_f from (14) assumes transparently the minimization of the simple quadratic criterion

$$J_f^{\text{LQR}} = \int_{t_f}^{\infty} \left(\|x(t)\|_{Q_f}^2 + \|u_f(t)\|_{R_f}^2\right) dt\,; \tag{18}$$

still, as no recommendations on the choice of the weighting matrices Q_f, R_f from (14)-(15) were given in [3], problem to which the authors brought contributions in [5]-[6], yet another extension of the result is being proposed in what follows.

IV. Approximate Model Matching II - Extended Hybrid Technique of MPIM with GLQR-EROMM Stabilization and Correction

For the novel solution that is to be developed in this section, Appendix I becomes a prerequisite.

1) MPIM with LQR-ROMM Stabilization and Correction: As no restrictions are indicated in the paper on MPIM [3], the authors have seen that the doublet of penalty matrices $Q_f^T = Q_f = \bar{Q}_f^T \bar{Q}_f \geq 0$ and $R_f^T = R_f > 0$ might be chosen as those from (46)-(49), while it does not matter much the quadratic criterion behind, as long as the solution stabilizes the impaired system.

In this case, it has been shown in [6] that, after the development of $\Delta K_f^{\text{PIM-LQR}}$ from (17), the MPIM-LQR control (16) provides to have somehow a bi-objective form as

$$K_f^{\text{MPIM-LQR}} = \tilde{\delta}_f^{\text{PIM-LQR}} K_f^{\text{PIM}} + (1 - \tilde{\delta}_f^{\text{PIM-LQR}}) K_f^{\text{CARE}}. \tag{19}$$

2) MPIM-GLQR-EROMM Extended Hybrid Technique: However, during the analysis, it has also been remarked that a more structured procedure than the simple LQR stabilization can be applied and hence the authors proposed to obtain the initial stabilizing post-fault control matrix as K_f^{GCARE} from (42), solution of GCARE_f from (43) associated with the faulty pair (A_f, B_f), after an appropriate choice of the full triplet of penalty matrices Q_f, R_f, S_f as in (46)-(49).

Then, similarly to (16), the hybrid control matrix becomes

$$K_f^{\text{MPIM-GLQR}} = K_f^{\text{GCARE}} + \tilde{\delta}_f^{\text{PIM-GLQR}} \Delta K_f^{\text{PIM-GLQR}}, \tag{20}$$

where the control accommodation difference $\Delta K_f^{\text{PIM-GLQR}}$ is obtained as

$$\Delta K_f^{\text{PIM-GLQR}} = B_f^{\dagger}\left(\hat{A}_f^{\text{GLQR}} - \hat{A}_n^*\right), \text{ for } \hat{A}_f^{\text{GLQR}} = A_f - B_f K_f^{\text{GCARE}}, \tag{21}$$

and thus, following the same logic as for the MPIM-LQR control, (20) gets the similar expression

$$K_f^{\text{MPIM-GLQR}} = \tilde{\delta}_f^{\text{PIM-GLQR}} K_f^{\text{PIM}} + (1 - \tilde{\delta}_f^{\text{PIM-GLQR}}) K_f^{\text{GCARE}}, \tag{22}$$

where $\delta_f^{\text{PIM-GLQR}}$ is calculated correspondingly.

Furthermore, remember that K_f^{GCARE} can be written as

$$K_f^{\text{GCARE}} = R_f^{-1} B_f^T X_f + R_f^{-1} S_f\,, \tag{23}$$

and then

$$K_f^{\text{GCARE}} = K_f^{\text{CARE}} + R_f^{-1} S_f\,, \tag{24}$$

and therefore $K_f^{\text{MPIM-GLQR}}$ from (22) changes its form depending on the case from which the classical / simplified / modified / extended pair of matrices (R_f, S_f) is chosen.

While the classical approach CROMM is similar to the simplified SROMM one and the extended approach EROMM to the modified MROMM one, it must be observed that the choice of the pair of matrices (R_f, S_f) that makes the difference in (24) is significant only in the cases classical versus extended.

As the results obtained in the situation of MPIM accommodation with the GLQR-CROMM stabilization were presented in [5]-[6], in this paper, the case of the novel MPIM accommodation with the GLQR-EROMM stabilization is developed.

Now, the second term of (24), $R_f^{-1} S_f$, after replacing again R_f and S_f from (49), becomes

$$R_f^{-1} S_f = \left(B_f^T B_f + I_m\right)^{-1} \left(B_f^T (A_f - \hat{A}_n^*) + K_n^*\right), \tag{25}$$

which can be split as

$$R_f^{-1} S_f = \underbrace{\left(B_f^T B_f + I_m\right)^{-1} B_f^T (A_f - \hat{A}_n^*)}_{\triangleq G} + \underbrace{\left(B_f^T B_f + I_m\right)^{-1} K_n^*}_{\triangleq H}; \tag{26}$$

in (26), denote by G and H its first and second terms, each one being treated separately in what follows.

First, for G, using one of the Searle identities [12]

$$(A + BB^T)^{-1} B = A^{-1} B (I + B^T A^{-1} B)^{-1}, \tag{27}$$

where A, B are two general matrices of appropriate dimensions and I is the identity matrix, the first term G of (26) writes as

$$G = \left(I_m + B_f^T B_f\right)^{-1} B_f^T (A_f - \hat{A}_n^*), \tag{28}$$

or

$$G = B_f^T \left(I_n + B_f B_f^T\right)^{-1} (A_f - \hat{A}_n^*). \tag{29}$$

Also, for H, using the Kailath invariant [13]

$$(A + BC)^{-1} = A^{-1} - A^{-1} B (I + C A^{-1} B)^{-1} C A^{-1}, \tag{30}$$

where A, B, C are three general matrices of appropriate dimensions and I is the identity matrix, the second term H of (26) writes as

$$H = \left(I_m + B_f^T B_f\right)^{-1} K_n^*, \tag{31}$$

or

$$H = I_m - B_f^T \left(I_n + B_f B_f^T\right)^{-1} B_f K_n^*. \tag{32}$$

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At this point, after replacing G and H from (29) and (32), (26) becomes

$$\begin{aligned} R_f^{-1}S_f =& I_m + \\ & B_f^T\left(I_n + B_f B_f^T\right)^{-1}(A_f - \hat{A}_n^*) - B_f^T\left(I_n + B_f B_f^T\right)^{-1} B_f K_n^* \\ =& I_m + B_f^T\left(I_n + B_f B_f^T\right)^{-1}(\hat{A}_f^* - \hat{A}_n^*), \end{aligned} \tag{33}$$

where by $\hat{A}_f^* = A_f - B_f K_n^*$ was denoted the closed-loop matrix that would have been obtained if the ideal control matrix K_n^* would have been applied directly to the impaired system (A_f, B_f).

Finally, after replacing (33) in (24), K_f^{GCARE} becomes

$$K_f^{\text{GCARE}} = K_f^{\text{CARE}} + I_m + B_f^T\left(I_n + B_f B_f^T\right)^{-1}(\hat{A}_f^* - \hat{A}_n^*) \tag{34}$$

and the MPIM-GLQR control (22) with EROMM weighting writes as

$$\begin{aligned} K_f^{\text{MPIM-GLQR}} =& (1 - \tilde{\delta}_f^{\text{PIM-GLQR}}) I_m + \\ & \tilde{\delta}_f^{\text{PIM-GLQR}} K_f^{\text{PIM}} + (1 - \tilde{\delta}_f^{\text{PIM-GLQR}}) K_f^{\text{CARE}} + \\ & (1 - \tilde{\delta}_f^{\text{PIM-GLQR}})\left(B_f^T\left(I_n + B_f B_f^T\right)^{-1}(\hat{A}_f^* - \hat{A}_n^*)\right). \end{aligned} \tag{35}$$

For this situation, the MPIM fault accommodation control redesign scheme with generalized GLQR-EROMM stabilization and control correction becomes a two step procedure as follows.

Algorithm 1: MPIM-GLQR-EROMM Scheme

(1-1) Stabilize the post-fault system using the generalized GLQR procedure of (43) with the weighting matrices Q_f, R_f, S_f as in (49), provide K_f^{GCARE} from (42).

(1-2) Make the appropriate control correction using the difference matrix $\Delta K_f^{\text{PIM-GLQR}}$ as in (21), provide $K_f^{\text{MPIM-GLQR}}$ from (20).

To highlight the novelty of the result developed in this paper, look again at the similarity between the controls computed with the simple MPIM-LQR (19) and the generalized MPIM-GLQR (35) and see that if the stability-bounds $\tilde{\delta}_f^{\text{PIM-G/LQR}}$ are much smaller than unity, their influence becomes negligible, such that (19) and (35) get the approximate form

$$K_f^{\text{MPIM-LQR}} \backsimeq K_f^{\text{CARE}}, \tag{36a}$$

$$K_f^{\text{MPIM-GLQR}} \backsimeq K_f^{\text{CARE}} + I_m + \left(B_f^T\left(I_n + B_f B_f^T\right)^{-1}(\hat{A}_f^* - \hat{A}_n^*)\right); \tag{36b}$$

this leads to the interpretation that, in the EROMM case, the PIM component has no influence in (36b) at all, which was not the case with the CROMM use presented in [5]-[6], as the corresponding control in that situation, under such hypothesis, was of the form

$$K_f^{\text{MPIM-LQR}} \backsimeq K_f^{\text{CARE}} + K_f^{\text{PIM}}; \tag{37}$$

in any case, even though having $\tilde{\delta}_f^{\text{PIM-G/LQR}} \ll 1$ is not always the case, still the further investigation of the differences between the two results is undergoing.

In the sequel, an illustrative application example on the redesign of the 2nd order controller for the short-period natural mode of behavior of the longitudinal model of a B747 civil aircraft is developed, with the aim of proving the advantages of using the generalized linear quadratic stabilization and modified pseudo-inverse method correction with post-fault penalty matrices provided by the extended robust optimal model matching approach, after a structural and an actuator fault.

V. Illustrative Application Example - Short-Period Mode of B747 Longitudinal Model

A. Flight Condition

Consider an aircraft Boeing 747 in gradual maneuvering without precision tracking (Class I cruise flight, Category B flight phase).

Assume a straight and constant level flight at 40000 ft fixed altitude and 0.8 Mach; the corresponding steady-state speed is 774 fps. (The CoG position is $(0.25, 0, 0)$ m.a.c. and the aircraft mass is 19792 $slug$.)

B. Nominal System

1) Open-Loop Model: The aircraft longitudinal short-period dynamics is described by the state vector $x = \begin{bmatrix} w & q \end{bmatrix}^T$, where w (ft/sec) represents the inertial velocity in the z- direction of the F_B body-axis reference frame, and q (rad/sec) represents the pitch rate. The control input δ_E (rad) is the elevator deflection.

The linearized model of the system is given by the nominal pair of system and control matrices (A_n, B_n) as (see [14])

$$A_n = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix},$$
$$B_n = \begin{bmatrix} -17.9416 & -1.1577 \end{bmatrix}^T.$$

2) Closed-Loop Reference Model: Consider the reference model $\hat{A}_n^* (\equiv \hat{A}_n)$ as

$$\hat{A}_n^* = \begin{bmatrix} -0.4573 & 731.9353 \\ -0.0103 & -3.1427 \end{bmatrix},$$

computed with the full state-feedback control gain

$$K_n^* = \begin{bmatrix} -0.0081 & -2.3445 \end{bmatrix},$$

which gives the stable spectrum

$$\Lambda(\hat{A}_n^*) = \{-1.8 \pm 2.4\imath\},$$

for the nominal closed-loop matrix $\hat{A}_n^* = A_n - B_n K_n^*$.

The reference model $\hat{A}_n^*$ provides ideal dynamic performance for this natural mode of the longitudinal motion of the aircraft, precisely for the short-period (SP) pair of complex-conjugate eigenvalues $\lambda_{1,2}^{SP} = -1.8 \pm 2.4\imath$ this gives the pair of natural undamped frequency (rad/sec) and damping coefficient $(\omega^{SP}, \zeta^{SP}) \approx (3, 0.6)$, which are perfect according to the thumb-print criterion.

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C. Impaired System

1) Post-Fault Model: Consider an actuator fault defined by the loss-of-effectiveness with 50% in the elevator deflection; that can be modeled in the post-fault system by multiplying the control matrix with a factor $(1-\tau_f)=0.5$, for $\tau_f=0.5$, i.e., $B'_f=(1-\tau_f)B_n$.

Also, assume a structural fault modeled in the system matrix as $A'_f=A_n+\Delta A_f\times I_2$, with $\Delta A_f=\sigma_n^{\max}\times 10^{-2}$, where $\sigma_n^{\max}\approx 774$ is the maximum singular value of the nominal matrix A_n.

Furthermore, consider also that the actual post-fault matrices provided by the FDIE module are $A_f=(1-\gamma_f)A'_f$ and $B_f=(1-\gamma_f)B'_f$, with $\gamma_f=0.1$, thus denoting a 10% structured identification error.

2) Pseudo-Inverse Solution: The Pseudo-Inverse Method (PIM) provides the control gain

$$K_f^{\text{PIM}}=\begin{bmatrix} -0.8810 & 4.2810 \end{bmatrix},$$

which is unacceptable, as it gives an unstable spectrum

$$\Lambda(\hat{A}_f^{\text{PIM}})=\{4.1912\pm 17.7454\imath\},$$

for the closed-loop matrix $\hat{A}_f^{\text{PIM}}=A_f-B_fK_f^{\text{PIM}}$.

3) Modified Pseudo-Inverse Solution: The Modified Pseudo-Inverse Method (MPIM) provides the control gain

$$K_f^{\text{MPIM}}=\begin{bmatrix} -0.1550 & 0.1550 \end{bmatrix},$$

which still is unacceptable, as it is not able to provide a stable spectrum

$$\Lambda(\hat{A}_f^{\text{MPIM}})=\{6.0472\pm 7.5233\imath\},$$

for the closed-loop matrix $\hat{A}_f^{\text{MPIM}}=A_f-B_fK_f^{\text{MPIM}}$.

This is due to the fact that matrix A_f is unstable, with the spectrum

$$\Lambda(A_f)=\{6.6325\pm 0.7901\imath\}.$$

D. Accommodated System - Extended ROMM Approach

1) MPIM-LQR Solution: The MPIM-LQR technique with stabilization and correction provides the control gain

$$K_{f_e}^{\text{MPIM-LQR}}=\begin{bmatrix} -2.1956 & -85.0870 \end{bmatrix},$$

which is acceptable, as it provides the stable spectrum

$$\Lambda(\hat{A}_{f_e}^{\text{MPIM-LQR}})=\{-11.4612,-37.3274\},$$

for the closed-loop matrix $\hat{A}_{f_e}^{\text{MPIM-LQR}}=A_f-B_fK_{f_e}^{\text{MPIM-LQR}}$, and the value of the overall cost (9) is

$$J_{f_e}^{\text{MPIM-LQR}}=\|\hat{A}_n^*-\hat{A}_{f_e}^{\text{MPIM-LQR}}\|_F=723.2104.$$

2) MPIM-GLQR Solution: Also, the MPIM-GLQR technique with stabilization and correction provides the control gain

$$K_{f_e}^{\text{MPIM-GLQR}}=\begin{bmatrix} -2.6444 & -79.4951 \end{bmatrix},$$

which is acceptable, as it provides the stable spectrum

$$\Lambda(\hat{A}_{f_e}^{\text{MPIM-GLQR}})=\{-19.6332,-29.8658\},$$

for the closed-loop matrix $\hat{A}_{f_e}^{\text{MPIM-GLQR}}=A_f-B_fK_{f_e}^{\text{MPIM-GLQR}}$, and the value of the overall cost (9) is

$$J_{f_e}^{\text{MPIM-GLQR}}=\|\hat{A}_n^*-\hat{A}_{f_e}^{\text{MPIM-GLQR}}\|_F=678.0477.$$

3) MPIM-GLQR vs. MPIM-LQR Control: Of course, looking at the first step of stabilization, the structural fault that changes matrix A_n into A_f can be assimilated with a structured perturbation, which is stabilizable according to the structured stability bounds within which the optimal solution exists, and the spectra in both G/LQR cases possess an intrinsic prescribed stability degree as well.

Hence, there is one advantage of the MPIM-G/LQR approaches in using such penalty matrices in that they provide real-parts of the closed-loop eigenvalues smaller than $-\alpha_f$ (where the minimal decay-rate $\alpha_f=1.5$ is chosen in relation with $\hat{\alpha}_n^*=1.8$ according to its definition, see later in the appendix), which makes both stand for good candidates if, in the post-fault operation, the prescribed degree-of-stability is also imposed for the performance to be followed as well (see also [15]).

However, the MPIM-GLQR approach provides some slightly smaller control gains and integral criterion value than the MPIM-LQR approach.

4) Closed-Loop Simulation: For the simulations, the full accommodation scenario is designed as with the following time evolution: from 0 to 5 *sec* there is the nominal (or ideal) evolution and until 6 *sec* the unstable unaccommodated evolution (impaired system with nominal control); thereafter, for 1 *sec* each (ideally considered so), the PIM and MPIM unstable regimes and then the MPIM-G/LQR-EROMM stable periods, with the stabilization and correction phases, are drawn one subsequent to the other. In both figures, the simulations for the short-period (SP) mode behavior (vertical inertial velocity and pitch rate) are shown differentiated as follows.

Due to limited space, in Fig. 1, only a truncated accommodation scenario is represented, that is the PIM and MPIM periods are eliminated, and this is to show that the magnitude of the instability decreases if the stabilization and correction technique is applied directly for the accommodation, without testing the classical approaches first. Anyway, in both cases, the accommodation is really abrupt, due to the fact that the quadratic stabilization is very fast and hence it recovers from instability very abruptly, that is why another approach like the step-by-step progressive accommodation of faults should be applied, solution which is to be investigated in a future series of other research papers.

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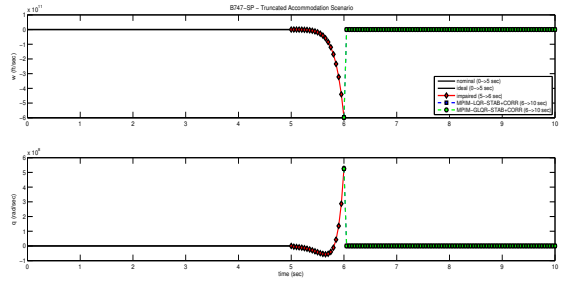

Fig. 1. MPIM-G/LQR-EROMM B747-SP - Truncated Accommodation Scenario

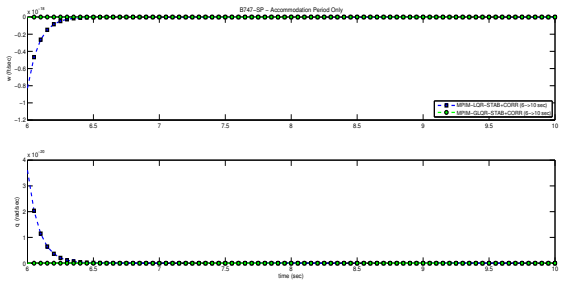

Fig. 2. MPIM-G/LQR-EROMM B747-SP - Accommodation Period Only

However, in Fig. 1, the stabilization and the correction phases are not easily to distinct one from the other due to the very small differences in the numerical values of the states in the stable phase compared to the big values that they have in the unstable phase - regarding that, in Fig. 2 only the accommodation period is represented, to show once again that the MPIM-GLQR is a slightly better follower than MPIM-LQR, the former being slightly faster convergent to the origin than the latter.

VI. Conclusion

This paper proposes an approximate solution that is used for the reconfiguration of an impaired system. The authors have developed a hybrid technique consisting of a two-stage controller redesign procedure with stabilization and correction, exploiting the synergy between the Modified Pseudo-Inverse Method (MPIM), Generalized Linear Quadratic Regulator (GLQR), and the new Extended Robust Optimal Model Matching (EROMM) approach, showing its specificity with regard to the previously proposed Classical ROMM technique. An application example for the fault tolerant control of the B747 short-period mode natural behavior is also provided for illustration.

Appendix I
Approximate Model Matching III - Robust Optimal Model Matching Approaches

The role of this section is to make a brief overview of some connected results that the authors are to propose in another paper, results which consist of other three novel types of similar solutions to the Classical Robust Optimal Model Matching (ROMM) technique.

These approaches, for which their corresponding controls preserve the prescribed stability degree and the robust stability margin of the original approach but the optimal cost is smaller, are called the Simplified, the Modified, and the Extended ROMM techniques. They are not given here on a sound basis, the technical details following in a more rigorous presentation.

However, from this perspective, and used in conjunction with the MPIM with stabilization and correction, for the first step of stabilization, two of the approaches will cost less, thus making them more promising for such hybrid technique.

Let $\hat{A}_n^*$ be the ideal nominal closed-loop matrix of the ideal model $\hat{M}_n^*$ from (3) with $\hat{\alpha}_n^* = |\lambda(\hat{A}_n^*)|$, where $|\lambda(\hat{A}_n^*)|$ is the maximum absolute value of the real-part of the eigenvalues of $\hat{A}_n^*$.

A. Robust Optimal Model Matching Quadratic Criteria

One of the classical solutions existing in the literature, used to reach this specific type of approximate model matching with particular robustness and optimality properties attached [16], considers that the post-fault state-feedback should be chosen so to minimize the performance index (see [4])

$$J_{f_c}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{\infty} e^{2\alpha_f t} \left(\|\dot{x}(t) - \hat{A}_n^* x(t)\|_2^2 + \|\hat{A}_n^* x(t)\|_2^2 \right) dt\,; \tag{38}$$

the exponent $\alpha_f \leq \hat{\alpha}_n^*$ is defined as the desired minimal decay-rate of the post-fault closed-loop system, namely the exponential convergence-rate of system state to the equilibrium point [7].

Recall the classical performance index J_{f_c} from (38) and consider its simplified version

$$J_{f_s}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{\infty} e^{2\alpha_f t} \left(\|\dot{x}(t) - \hat{A}_n^* x(t)\|_2^2 \right) dt\,; \tag{39}$$

in this approach, the second term from the original integral criterion has been removed to allow the designer focus just on the minimization of the norm between the actual impaired trajectory and the nominal ideal one.

Moreover, the term $\hat{A}_n^*$ could be called even detracting, because it does not actually lead to a convergence between the two models to be matched, but it causes a difference between the original and the matched model even the systems are identical.

Then, consider the modified version of J_{f_c}, that is

$$J_{f_m}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{\infty} e^{2\alpha_f t} \left(\|\dot{x}(t) - \hat{A}_n^* x(t)\|_2^2 + \|u_f(t) - u_n^*(t)\|_2^2 \right) dt\,; \tag{40}$$

in this approach, the second term in the integral criterion is introduced for a more realistic point of view regarding the energy consumption to attain the fault tolerant control objective.

More specifically, this implies that the robust optimal model matching action is performed with some control effort but not much greater than that used in the nominal operation.

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Furthermore, consider also the extended version of J_{f_c}, that is

$$J_{f_e}(\alpha_f, \hat{A}_n^*) = \int_{t_f}^{\infty} e^{2\alpha_f t} \left(\|\dot{x}(t) - \hat{A}_n^* x(t)\|_2^2 + \|u_f(t) - u_n^*(t)\|_2^2 + \|\hat{A}_n^* x(t)\|_2^2 \right) dt\,; \tag{41}$$

in this approach, all the integrand terms identified up to now are included in the optimization process.

The last term comes from the classical approach, which still assumes and imposes having minimum variance for the ideal trajectory.

B. Robust Optimal Model Matching Control Solutions

It has been shown that, in any of the previously introduced cases, the post-fault state-feedback $u_{f_k}(t)$, chosen to minimize $J_{f_k}(\alpha_f, \hat{A}_n^*)$ from (38)-(41), subject to (A_f, B_f), is the generalized optimal control law

$$u_{f_k}(t) = -K_{f_k} x(t)\,, \text{ with } K_{f_k} = R_{f_k}^{-1}(B_f^T X_{f_k} + S_{f_k})\,, \tag{42}$$

for X_{f_k} the solution to the Generalized Continuous Algebraic Riccati Equation (GCARE) from the classical / simplified / modified / extended criterion case

$$GCARE_{f_k} : \tilde{A}_{f_k}^T X_{f_k} + X_{f_k}\tilde{A}_{f_k} - X_{f_k} B_f R_{f_k}^{-1} B_f^T X_{f_k} + \tilde{Q}_{f_k} = 0\,, \tag{43}$$

with the buffer matrices

$$\tilde{A}_{f_k} = A_f + \alpha_f I_n - B_f R_{f_k}^{-1} S_{f_k}\,,\ \tilde{Q}_{f_k} = Q_{f_k} - S_{f_k}^T R_{f_k}^{-1} S_{f_k}\,, \tag{44}$$

for $k \in \{c, s, m, e\}$, where the legend of the indexes is as follows: $c =$ classical, $s =$ simplified, $m =$ modified, $e =$ extended.

The required post-fault weighting matrices $Q_{f_k}, R_{f_k}, S_{f_k}$ from (42)-(44) are identified after the reduction of the performance integrals (38)-(41) to the corresponding generalized-form

$$J_f^{\text{GLQR}} = \int_{t_f}^{\infty} e^{2\alpha_f t} \left(\|x(t)\|_{Q_f}^2 + \|u_f(t)\|_{R_f}^2 + 2u_f(t)^T S_f x(t) \right) dt\,, \tag{45}$$

as follows: for the Classical Robust Optimal Model Matching (CROMM) procedure as

$$Q_{f_c} = \left(A_f - \hat{A}_n^*\right)^T \left(A_f - \hat{A}_n^*\right) + \hat{A}_n^{*T} \hat{A}_n^*, \tag{46a}$$

$$R_{f_c} = B_f^T B_f\,,\ S_{f_c} = B_f^T (A_f - \hat{A}_n^*)\,, \tag{46b}$$

for the Simplified (SROMM) procedure as

$$Q_{f_s} = Q_{f_c} - \hat{A}_n^{*T} \hat{A}_n^*\,,\ R_{f_s} = R_{f_c}\,,\ S_{f_s} = S_{f_c}\,, \tag{47}$$

for the Modified (MROMM) procedure as

$$Q_{f_m} = Q_{f_c} - \hat{A}_n^{*T} \hat{A}_n^* + K_n^{*T} K_n^*\,,\ R_{f_m} = R_{f_c} + I_m\,,\ S_{f_m} = S_{f_c} + K_n^*\,, \tag{48}$$

and for the Extended (EROMM) procedure as

$$Q_{f_e} = Q_{f_c} + K_n^{*T} K_n^*\,,\ R_{f_e} = R_{f_c} + I_m\,,\ S_{f_e} = S_{f_c} + K_n^*\,, \tag{49}$$

for the squared Q_f- and R_f- state- and control- vectors norms, i.e., $\|x(t)\|_{Q_f}^2 = x(t)^T Q_f x(t)$ and $\|u_f(t)\|_{R_f}^2 = u_f(t)^T R_f u_f(t)$, and S_f the state-control coupling matrix.

As the results obtained for the Classical ROMM approach using the pair (R_{f_c}, S_{f_c}) (similar to the SROMM, as $R_{f_s} = R_{f_c}$ and $S_{f_s} = S_{f_c}$) were presented in [5]-[6], the other novel and different situation for the Extended ROMM approach using the pair (R_{f_e}, S_{f_e}) is developed (just the EROMM is chosen for the analysis in the paper, because it is similar to the MROMM, as $R_{f_m} = R_{f_e}$ and $S_{f_m} = S_{f_e}$).

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REFERENCES

[1] M. Verhaegen, S. Kanev, R. Hallouzi, C. Jones, J. Maciejowski, and H. Smaili, "Fault tolerant flight control - A survey," in *Fault Tolerant Flight Control*, ser. Lecture Notes in Control and Information Sciences, C. Edwards, T. Lombaerts, and H. Smaili, Eds. Springer, 2010, vol. 399, ch. 2, pp. 47–89.

[2] C. Belcastro, "Aircraft loss of control: Analysis and requirements for future safety-critical systems and their validation," in *Proceedings of the 8th Asian Control Conference*, Kaohsiung, Taiwan, 2011, pp. 399–406.

[3] Z. Gao and P. Antsaklis, "Stability of the pseudo-inverse method for reconfigurable control systems," *International Journal of Control*, vol. 53, no. 3, pp. 717–729, 1991.

[4] M. Zohdy, N. Loh, and A.-A. Abdul-Wahab, "A robust optimal model matching control," *IEEE Transactions on Automatic Control*, vol. 32, no. 5, pp. 410–414, 1987.

[5] B. Ciubotaru and M. Staroswiecki, "Extension of modified pseudo-inverse method with generalized linear quadratic stabilization," in *Proceedings of the 29th American Control Conference*, Baltimore, Maryland, USA, 2010, pp. 6222–6224.

[6] B. Ciubotaru, M. Staroswiecki, and N. Christov, "Modified pseudo-inverse method with generalized linear quadratic regulator for fault tolerant model matching with prescribed stability degree," in *Proceedings of the 50th IEEE Conference on Decision and Control, and the 11th European Control Conference*, Orlando, Florida, USA, 2011, pp. 1583–1588.

[7] B. Anderson and J. Moore, "Linear system optimisation with prescribed degree of stability," *Proceedings of the IEEE*, vol. 116, no. 12, pp. 2083–2087, 1969.

[8] H. Erzberger, "Analysis and design of model following control systems by state space techniques," in *Proceedings of the Joint Automatic Control Conference*, Ann Arbor, Michigan, USA, 1968, pp. 572–581.

[9] B. Ciubotaru, M. Staroswiecki, and C. Christophe, "Fault tolerant control of the Boeing 747 short-period mode using the admissible model matching technique," in *Proceedings of the 6th IFAC Symposium on Fault Detection, Supervision and Safety of Technical Processes*, Beijing, P.R. China, 2006, pp. 819–824.

[10] R. Yedavalli, "Improved measures of stability robustness for linear state space models," *IEEE Transactions on Automatic Control*, vol. 30, no. 6, pp. 577–579, 1985.

[11] K. Zhou and P. Khargonekar, "Stability robustness bounds for linear state-space models with structured uncertainty," *IEEE Transactions on Automatic Control*, vol. 32, no. 7, pp. 621–623, 1987.

[12] S. Searle, *Matrix Algebra Useful for Statistics*. Wiley, 2006.

[13] T. Kailath and A. Sayed, Eds., *Fast Reliable Algorithms for Matrices with Structure*. SIAM, 1987.

[14] B. Etkin and L. Reid, *Dynamics of Flight: Stability and Control*, 3rd ed. John Wiley & Sons, 1996.

[15] A.-A. Abdul-Wahab, "Robustness measure bounds for optimal model matching control designs," *IEEE Transactions on Automatic Control*, vol. 33, no. 12, pp. 1178–1180, 1988.

[16] R. Patel, M. Toda, and B. Sridhar, "Robustness of linear quadratic state feedback in the presence of system uncertainty," *IEEE Transactions on Automatic Control*, vol. 22, no. 6, pp. 945–949, 1977.

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